
Jonathan Zajac | Ph.D. Candidate | zajac028@umn.edu | jonathan-zajac@github.io
Department of Chemistry | University of Minnesota, Minneapolis, MN

Enthusiastic, consistent, and creative computational molecular scientist, specializing in investigating complex biophysical problems. Proficient in biomolecular simulations and experience working with systems related to protein folding, aggregation, virus capsid stability, and GPCR pharmacology. Committed to impactful advancements in biology and therapeutics through collaborative research.

EDUCATION

Ph.D.	University of Minnesota, Minneapolis, MN	2026 (est.)	Chemistry
M.S.	University of Minnesota, Minneapolis, MN	2023	Chemistry
B.S.	University of Wisconsin, Eau Claire, WI	2020	Biochemistry, <i>Major</i> Neuroscience, <i>Minor</i>

RESEARCH EXPERIENCE

2021 – Present **University of Minnesota – Twin Cities, Dept. of Chemistry**
Advisor: Dr. Sapna Sarupria

Investigating Molecular Determinants of Excipient-Mediated Biomolecular Stability

Established a concentration-dependent “flipping” mechanism of arginine on hydrophobic interactions from a direct to indirect stabilizer. Identified molecular determinants for co-excipient synergy related to hydrophobic polymer folding in binary amino acid solutions. Developed a computationally efficient simulation strategy for investigating virus-solvent interactions at atomistic resolution. Demonstrated a unifying framework for excipient effects on the temperature dependence of protein folding/unfolding through shape complementarity. Provided mechanistic insights through molecular simulations on novel agonists/antagonists targeting melanocortin receptors.

2019 – 2020 **University of Wisconsin – Eau Claire, Dept. of Psychology**
Advisor: Dr. David Jewett

Identified and assessed individual differences in stimulus control among rats trained to discriminate between naltrexone and saline. Developed skills in operant conditioning and data analysis. Learned ethical and proper animal care and use in compliance with the Institutional Animal Care and Use Committee.

2017 – 2019 **University of Wisconsin – Eau Claire, Dept. of Chemistry**
Advisor: Dr. Sudeep Bhattacharyay

Performed hybrid quantum mechanical/molecular mechanical simulations to assess ligand binding, protein function, and dynamics. Developed methods to identify optimal binding orientations for inhibitors of NAD(P)H:quinone oxidoreductases using geometric and energetic methods. Utilized DynaMut, CHARMM, and VMD to predict the effects of distal mutation on the flexibility and stability of prolyl-tRNA synthetase.

TEACHING & MENTORING EXPERIENCE

2024 – Present **Research Mentor**

Led a team of six junior researchers in the design and execution of a computational project on *de novo* protein design and engineering. Group composition included 1 master's student, 2 undergraduate students, and 3 high school students, spanning 5 different institutions. All students have presented their work at science fairs and research exhibitions. One student earned first place for the Medtronic Innovation Award and Regeneron Award.

2021 - 2022 **Teaching Assistant, University of Minnesota, Chemistry**

Lectured and oversaw two general chemistry labs, each comprised of 26 undergraduate students. Held office hours and helped design course materials for thermodynamics.

2019 - 2020 **Tutor, University of Wisconsin – Eau Claire, Chemistry**

2019 - 2020 **Teaching Assistant, University of Wisconsin – Eau Claire, Psychology**

HONORS & AWARDS

- 2026 **Anton 3 Allocation**, awarded 5 machine-days for atomistic simulations of empty and full AAV8 viral capsids. (MCB250009P)
- 2026 **NSF Discover ACCESS Grant**, awarded 1,500,000 CPU hours for simulations of excipient-mediated protein aggregation suppression. (BIO260133)
- 2025 **Graduate Student Award in Computational Molecular Science and Engineering (CoMSEF)**, for excellence in research by graduate students and significant contributions to research in computational molecular science and engineering. Recognized at the AIChE 2025 Annual Meeting.
- 2025 **Doctoral Dissertation Fellowship**, awarded a \$25,000 stipend, tuition stipend, and \$1,000 travel award. Awarded out of 195 doctoral candidate applicants at the University of Minnesota.
- 2025 **NSF Maximize ACCESS Grant**, awarded 14,000,000 CPU hours for simulations towards the design of stable virus and virus-like particle formulations. (CHE240118)
- 2025 **i-CoMSE Travel Award**, awarded a \$500 travel award to attend the 10th i-CoMSE workshop on QM/MM and AIMD, hosted at Oklahoma State University.
- 2024 **NSF Accelerate ACCESS Grant**, awarded 3,000,000 CPU hours for the development of a novel computational approach for *de novo* enzyme design. (BIO240236)
- 2024 **NSF Accelerate ACCESS Grant**, awarded 3,000,000 CPU hours for simulations towards the design of stable virus formulations. (CHE240042)
- 2024 **Equity Certificate Hosted Online (ECHO)**, Office for Equity and Diversity (OED), University of Minnesota – Twin Cities
- 2024 **Grand Prize (\$2000 Travel Award)**, Research Computing Exhibition, University of Minnesota – Twin Cities
- 2023 **NSF Explore ACCESS Grant**, awarded 400,000 CPU hours for the development of a novel approach for atomistic capsid simulations. (CHE240042)
- 2023 **Top Poster (\$1000 Travel Award)**, Research Computing Exhibition, University of Minnesota – Twin Cities
- 2021 **Thank a Teacher Certificate**, received Fall 2021 for Outstanding Teaching Assistant through the Center for Educational Innovation at the University of Minnesota.
- 2020 **Departmental Honors**, Dept. of Chemistry, University of Wisconsin – Eau Claire
- 2016 **Honors Scholarship**, University of Wisconsin – Eau Claire

PUBLICATIONS | Peer-Reviewed Journal Articles

1. **J. W. P. Zajac**, I. Tohidian, P. Muralikrishnan, S. L. Perry, C. L. Heldt, and S. Sarupria. "Cracking the Capsid Code: A Computationally Feasible Approach for Investigating Virus–Excipient Interactions in Biologics Design." *Journal of Chemical Theory and Computation*, 2026, 22(5), 2635–51. <https://doi.org/10.1021/acs.jctc.5c01810>.
2. **J. W. P. Zajac**, P. Muralikrishnan, C. L. Heldt, S. L. Perry, S. Sarupria, "Towards Stable Biologics: Understanding Co-Excipient Effects on Hydrophobic Interactions and Solvent Network Integrity",

Molecular Systems Design & Engineering, 2025, 10, 432-446. doi: 10.1039/D4ME00201F
(Selected for front cover)

3. **J. W. P. Zajac**, P. Muralikrishnan, I. Tohidian, X. Zeng, C. L. Heldt, S. L. Perry, S. Sarupria, "Flipping Out: Role of Arginine in Hydrophobic Polymer Collapse". Chemical Science, 2025, 16, 6780-6792. doi: 10.1039/d4sc08672d
4. P. Muralikrishnan, **J. W. P. Zajac**, C. L. Heldt, S. L. Perry, S. Sarupria, "Thermodynamic Basis of Sugar-Dependent Polymer Stabilization: Informing Biologic Formulation Design", Journal of Physical Chemistry B, 2025. doi: 10.1021/acs.jpcc.5c06203
5. N. Weirath, **J. W. P. Zajac**, H. Donow, T. Lavoie, C. Pinilla, R. Santos, R. Prajapati, R. Speth, M. Ericson, S. Sarupria, M. Giulianotti, C. Haskell-Luevano, "N-branched tri-cyclic guanidines as novel melanocortin-3 receptor agonists and melanocortin-4 receptor antagonists". Journal of Medicinal Chemistry, 2025, 68, 2504-2527. doi: 10.1021/acs.jmedchem.4c01556
6. S. Dasetty, **J. W. P. Zajac**, and S. Sarupria, "Exploitation of active site flexibility-low temperature activity relation for engineering broad range temperature active enzymes", Molecular Systems Design and Engineering (Emerging Investigator series) 2023, 8, 1355-1370. doi: 10.1039/D3ME00013C (Selected for front cover)
7. S. Suhail, **J. W. P. Zajac**, C. Fossum, H. Lowater, C. McCracken, N. Severson, B. Laatsch, A. Narkiewicz-Jodko, B. Johnson, J. Liebau, S. Bhattacharyya, S. Hati, "Role of Oxidative Stress on SARS-CoV (SARS) and SARS-CoV-2 (COVID-19) Infection: A Review", Protein Journal, 2020, 39(6), 644-656. doi: 10.1007/s10930-020-09935-8
8. **J. W. P. Zajac**, H. Anderson, L. Adams, D. Wangmo, S. Suhail, A. Almen, L. Berns, B. Coerber, L. Dawson, A. Hunger, J. Jehn., J. Johnson, N. Plack, S. Strasser, M. Williams, S. Bhattacharyya, S. Hati, "Effects of Distal Mutations on Prolyl-Adenylate Formation of Escherichia coli Prolyl-tRNA Synthetase", Protein Journal, 2020 39(5), 542-553. doi: 10.1007/s10930-020-09910-3

PUBLICATIONS | Preprints & Manuscripts in Preparation

8. **J. W. P. Zajac**, P. Muralikrishnan, X. Zeng, S. L. Perry, C. L. Heldt, S. Sarupria. "Protein-Solvent Shape Complementarity as a Unifying Phenomenon in Excipient-Driven Formulation Stability" [Manuscript in preparation, anticipated submission April 2026]
9. I. Tohidian, X. Zeng, **J. W. P. Zajac**, R. Chaudhari, S. Sarupria, S. Perry, C. L. Heldt. "Mechanistic Understanding of the Effect of Excipients on the Thermal Stability of Biological Molecules" [Manuscript in preparation, anticipated submission May 2026]
10. X. Zeng, **J. W. P. Zajac**, A. Sathyavageswaran, C. L. Heldt, S. Sarupria, S. L. Perry. "Exploring the Mutual Effects of Excipient and Proteins in Complex Coacervation" [Manuscript in preparation, anticipated submission May 2026]
11. **J. W. P. Zajac**, M. Ericson, C. Haskell-Luevano, S. Sarupria. "Agonist Binding Enriches Hydration of Extracellular-Intracellular Communication Pathways in Melanocortin Receptors" [Manuscript in preparation, anticipated submission June 2026]

PRESENTATIONS | Professional Meetings & Conferences

1. **J. W. P. Zajac**, P. Muralikrishnan, I. Tohidian, X. Zeng, S. Heldt, S. Perry, S. Sarupria, "Blueprinting Future-Proof Vaccines: Enhancing Formulation Stability through Molecular Simulations and Network Theory", November 2025, poster presented at American Institute of Chemical Engineers (AIChE) 2025 Annual Meeting, Boston, MA.
2. **J. W. P. Zajac**, P. Muralikrishnan, C. Heldt, S. Perry, S. Sarupria, "Probing Excipient-Driven Stability in Vaccines via Molecular Dynamics Simulations", June 2025, oral presentation at the 6th Modeling Workshop, hosted by GSK in Collegeville, PA.

3. **J. W. P. Zajac**, P. Muralikrishnan, C. Heldt, S. Perry, S. Sarupria, "Excipient architects: Redesigning solvent networks for improved protein temperature stability in liquid formulations", March 2025, oral presentation at American Chemical Society annual meeting in San Diego, CA.
4. **J. W. P. Zajac**, P. Muralikrishnan, S. Sarupria, "Insights Into Excipient-Driven Stabilization of Viral Capsid Fragments Using Molecular Simulations", June 2024, oral presentation at American Society of Virology 2024 annual meeting in Columbus, OH.
5. **J. W. P. Zajac**, P. Muralikrishnan, S. Sarupria, "Designing Protein Stabilizers: Small Molecule Effects on Protein Folding are Driven by Direct Interactions and Solvent Rearrangement", March 2024, oral presentation at American Physical Society annual meeting in Minneapolis, MN.

PRESENTATIONS | Posters & Exhibitions

6. **J. W. P. Zajac**, P. Muralikrishnan, S. Sarupria, "Navigating Complex Design Space in Modeling Temperature-Stable Therapeutics", Oct. 2024, poster presented at the NIH Chemical-Biology Interface Training Grant Symposium, University of Minnesota.
7. **J. W. P. Zajac**, S. Sarupria, "Balancing Excipient Mechanisms in the Design of Temperature-Stable Biologics", June 2024, oral presentation at the 2024 Graduate Student Research Symposium, University of Minnesota.
8. **J. W. P. Zajac**, P. Muralikrishnan, S. Sarupria, "Navigating Complex Design Space in Modeling Temperature-Stable Therapeutics", May 2024, poster presented at the Minnesota Supercomputing Institute / Research Computing Exhibition, University of Minnesota. **(Awarded grand prize)**
9. **J. W. P. Zajac**, P. Muralikrishnan, S. Sarupria, "Investigating the Effects of Arginine on Hydrophobic Interactions: Insights Towards Temperature-Stable Vaccine Design", May 2023, poster presented at the Minnesota Supercomputing Institute / Research Computing Exhibition, University of Minnesota. **(Awarded travel award)**
10. **J. W. P. Zajac**, P. Muralikrishnan, S. Sarupria, "Investigating the Effects of Arginine on Hydrophobic Interactions: Insights Towards Temperature-Stable Vaccine Design", May 2023, poster presented at the NIH Chemical-Biology Interface Training Grant Symposium, University of Minnesota.
11. **J. W. P. Zajac**, P. Muralikrishnan, C. Heldt, S. Perry, S. Sarupria, "Towards Machine Learning-Enabled Discovery of Thermostable Vaccines", Mar. 2023, poster presented at the Peter O. Stahl Advanced Design Forum, University of Minnesota.
12. **J. W. P. Zajac**, A. Vigil, S. Petit, E. Lucas, D. Jewett, "Stimulus Control Differences in Rats Given Chronic Sucrose Access and Trained to Discriminate Between Naltrexone and Saline", 2020, oral presentation at the Neuroscience Undergraduate Research Virtual Symposium (NURVS).
13. **J. W. P. Zajac**, A. Vigil, S. Petit, E. Lucas, D. Jewett, "Individual Differences in Stimulus Control Among Rats Trained to Discriminate Between Naltrexone and Saline", 2020, oral presentation at MidBrains Midwest Regional Neuroscience Conference, hosted by Augustana College.
14. **J. W. P. Zajac**, H. Stutt, H. Dorn, A. Vigil, A. Petrey, S. Petit, M. Marek, D. Jewett, "Individual Differences in Stimulus Control in Rats with Chronic-Intermittent Sucrose Access Trained to Discriminate Between Naltrexone and Saline", 2020, Abstract, FASEB Journal, 34(51). doi: 10.1096/fasebj.2020.34.s1.04224
15. **J. W. P. Zajac**, H. Stutt, H. Dorn, A. Petrey, C. Herzberg, M. Marek, D. Jewett, "Are Naltrexone's Discriminative Stimulus Effects Mediated by Kappa-Opioid Receptors in Rats with Chronic, Intermittent Sucrose Access?", 2019, poster presented at the annual Society for Neuroscience conference in Chicago, IL.
16. **J. W. P. Zajac**, N. Fruit, E. Schulz, O. Vile, H. Stutt, H. Dorn, D. Jewett, "Effects of the Kappa Agonist U69,593 on Naltrexone's Discriminative Stimulus Effect in Subjects Given Chronic,

Intermittent Sucrose Access”, 2019, poster presented at Celebration of Excellence in Research + Creativity (CERCA) at the University of Wisconsin – Eau Claire.

17. **J. W. P. Zajac**, S. Bhattacharyay, “Docking and Energetic Studies of Quinones and Other Aromatic Ligands in Quinone Reductases”, 2018, poster presented at the annual Molecular Education and Research Consortium in Undergraduate Computational Chemistry (MERCURY) at Furman University in Greenville, SC.
18. **J. W. P. Zajac**, S. Bhattacharyay, “Calculation of Binding Free Energies of NAD(P)H:Quinone Oxidoreductase 1 Inhibitors”, 2018, poster presented at Celebration of Excellence in Research + Creativity (CERCA) at the University of Wisconsin – Eau Claire.

SKILLS | Key Research Skills

- **Programming Languages:** Bash, Python, Tcl
- **Version Control and Documentation:** Git, LaTeX
- **Molecular Simulation Engines:** GROMACS, CHARMM, openMM
- **Molecular Modeling Software:** VMD, Avogadro, PyMol, Chimera
- **Molecular Docking Software:** AutoDock, DiffDock, SwissDock
- **Protein Design Software:** RFdiffusion, LigandMPNN, AlphaFold, Chai-1, ColabFold
- **Molecular Analysis Tools:** MDAnalysis, MDTraj, DynaMut, CNAalysis